PS- I WEEKLY REPORT

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1. Executive Summary

This week, the team successfully completed data preprocessing and built, trained, and evaluated multiple machine learning classification models on the 2021–2023 NHANES CKD (Chronic Kidney Disease) dataset sourced from Kaggle. The pipeline included data cleaning, removal of irrelevant multi-class columns, encoding, feature scaling, and an 80:20 train-test split before model training. A total of eight algorithms were implemented and benchmarked. The results are highly promising, with several models achieving perfect classification accuracy. However, given the nature of the dataset, these results should be interpreted with caution and validated further. Model performance was evaluated using five metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC..

2. Dataset Overview

Parameter	Details
Source	Kaggle (12000 RECORDS)
Dataset Name	2021–2023 NHANES CKD Dataset
Task Type	Binary Classification
Training Set	80%
Testing Set	20%

3. Data Preprocessing Pipeline

Before model training, a structured preprocessing pipeline was applied to ensure data quality and consistency.

3.1 Data Cleaning

- Checked for and handled missing/null values
- Removed duplicate records where applicable
- Verified data types and corrected inconsistencies

3.2 Column Removal — Multi-class Features

During the preprocessing stage, it was identified that two columns — **CKD_STAGE** and **EGFR** — were associated with multi-class classification labels. Since the objective of this project is strictly **binary classification** (CKD present vs. not present), these two columns were removed from the dataset to avoid label leakage, target confusion, and model bias.

3.3 Encoding

- **Label Encoding / One-Hot Encoding** applied to all categorical features to convert them into numerical format suitable for ML algorithms
- Target variable was label-encoded for classification

3.3 Feature Scaling

- **Standard Scaling (Z-score normalization)** applied to continuous numerical features
- Ensured all features are on a uniform scale, particularly important for distance-based models like KNN and SVM

3.4 Train-Test Split

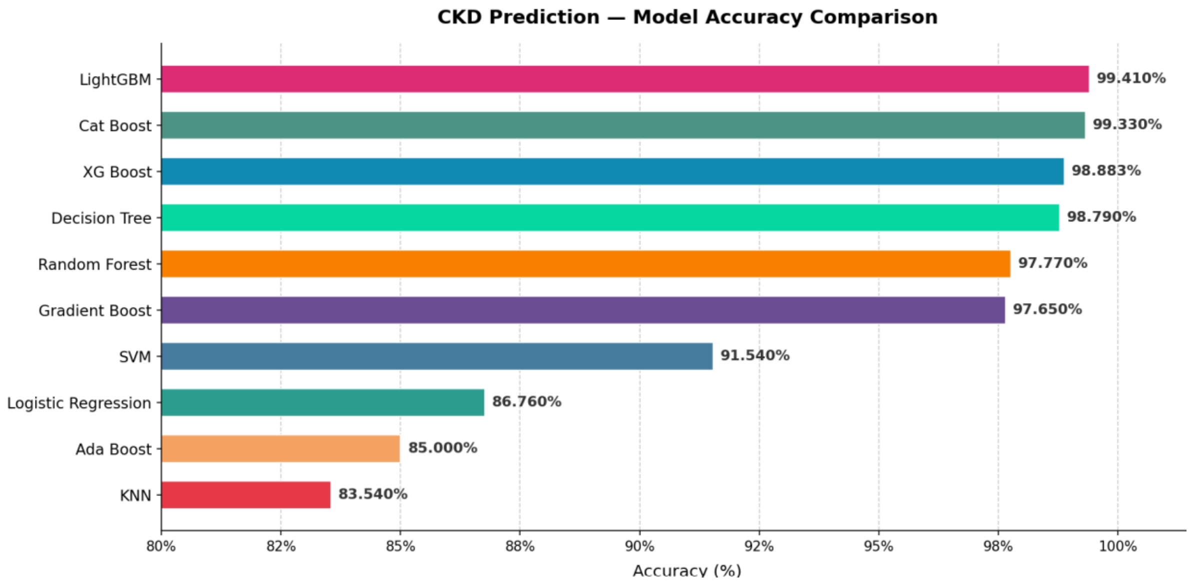
- Dataset split in an **80:20 ratio**

4. Models Implemented & Accuracy Results

Eight classifiers were trained on the preprocessed data. The table below summarizes their accuracy on the test set:

Model	Accuracy
AdaBoost	85%
Decision Tree (DT)	98.79%

Model	Accuracy
Gradient Boosting	97.65%
K-Nearest Neighbors (KNN)	83.54%
Logistic Regression	86.76%
Random Forest	97.77%
Support Vector Machine (SVM)	91.54%
XGBoost	98.883%
LightGBM	99.41%
Car Boost	99.33%



5. Evaluation Metrics

Each model was assessed using the following classification metrics:

Metrics Used:

- **Accuracy** — Overall correct predictions out of total samples
- **Precision** — Of all positive predictions, how many were actually positive
- **Recall** — Of all actual positives, how many were correctly identified
- **F1-Score** — Harmonic mean of Precision and Recall
- **ROC-AUC** — Area under the Receiver Operating Characteristic curve; measures separability

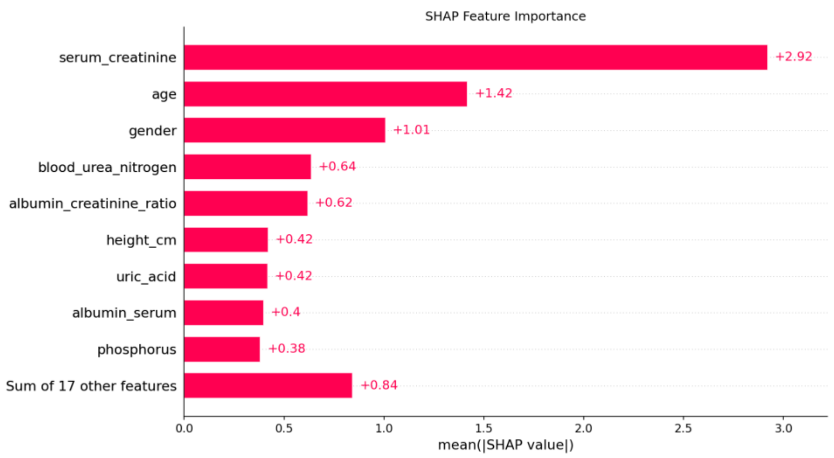
Rank	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
1	LightGBM	99.41%	99.76%	99.46%	99.58%	99.99%
2	Cat Boost	99.33%	99.70%	99.34%	99.52%	99.98%
1	XGBoost	98.83%	99.88%	98.44%	99.15%	99.98%
2	Decision Tree	98.79%	99.52%	98.74%	99.13%	99.50%
3	Random Forest	97.70%	98.85%	97.84%	98.34%	99.81%
4	Gradient Boosting	97.65%	98.91%	97.72%	98.31%	99.85%
5	SVM	91.54%	97.72%	89.99%	93.70%	97.47%
6	Logistic Regression	86.76%	87.43%	94.66%	90.90%	90.21%
7	AdaBoost	85%	88.60%	91.31%	89.93%	92.92%
8	KNN	83.54%	84.97%	92.87%	88.74%	89.46%

IMPLEMENTATION OF XAI (SHAP):

1. SHAP Feature Importance Plot:

Rank	Feature	Mean SHAP
1	Serum Creatinine	2.92
2	Age	1.42
3	Gender	1.01
4	Blood Urea Nitrogen	0.64
5	Albumin-Creatinine Ratio	0.62
6	Height	0.42
7	Uric Acid	0.42
8	Serum Albumin	0.40
9	Phosphorus	0.38

SHAP feature importance analysis identified Serum Creatinine as the most influential predictor of CKD, followed by Age, Gender, Blood Urea Nitrogen, and Albumin-Creatinine Ratio. These findings align with established clinical knowledge regarding renal dysfunction and CKD progression.



2. SHAP Summary Plot (Beeswarm):

This graph tells us how feature values affect predictions.

Interpretation

- Higher serum creatinine strongly increases CKD probability.
- Lower serum creatinine decreases CKD probability.
- This is exactly what nephrologists would expect

Age

- High age values (red) are mostly on the right.
- Low age values (blue) are on the left.

Interpretation

Older age increases CKD risk.

Younger age decreases CKD risk.

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Albumin-Creatinine Ratio

- High values lie far to the right.
- Low values stay around zero or negative.

Interpretation

Higher urine protein leakage strongly pushes the model toward CKD.

Blood Urea Nitrogen

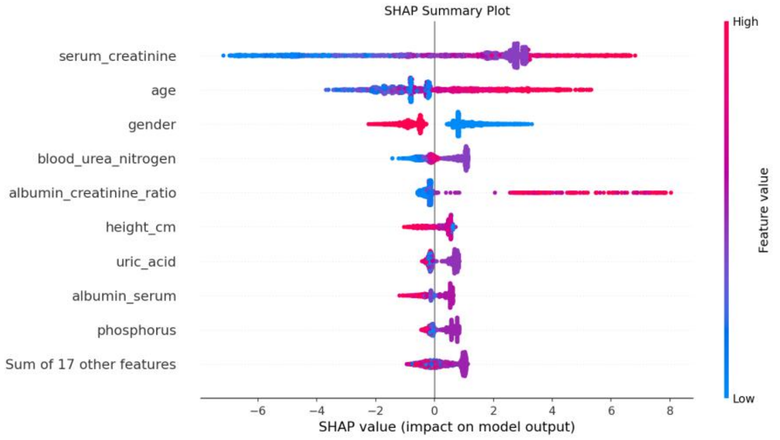
Higher BUN contributes positively toward CKD prediction.

This is clinically meaningful.

Uric Acid

Higher uric acid levels tend to push predictions toward CKD.

This relationship is often reported in kidney disease studies.



3. Waterfall Plot (Individual Patient):

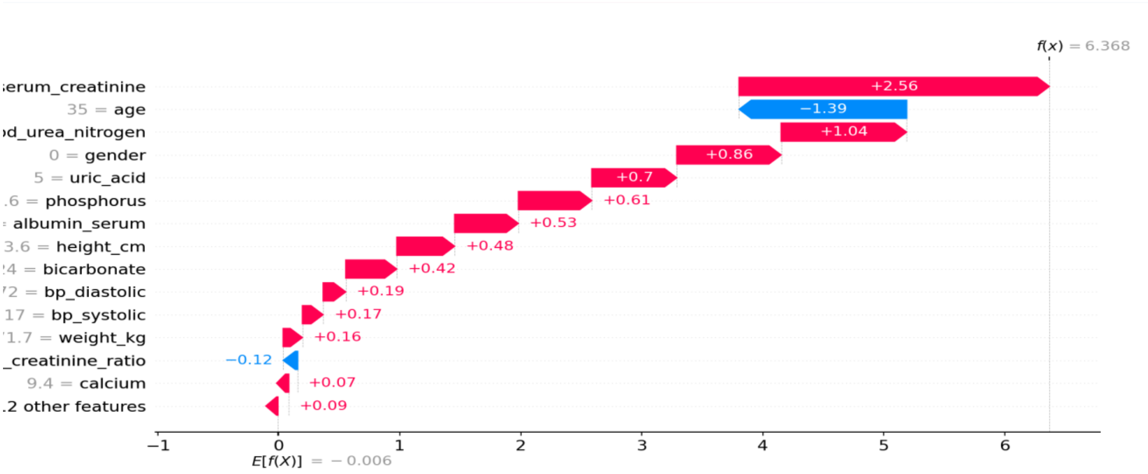
This plot explains one patient's prediction.

Feature	Contribution
Serum Creatinine	+2.56
Age	+1.39
Blood Urea Nitrogen	+1.04
Gender	+0.86
Uric Acid	+0.70

Interpretation

This patient is predicted as CKD with very high confidence because:

- Serum creatinine is elevated
- Blood urea nitrogen is elevated
- Uric acid is elevated
- Age contributes significantly



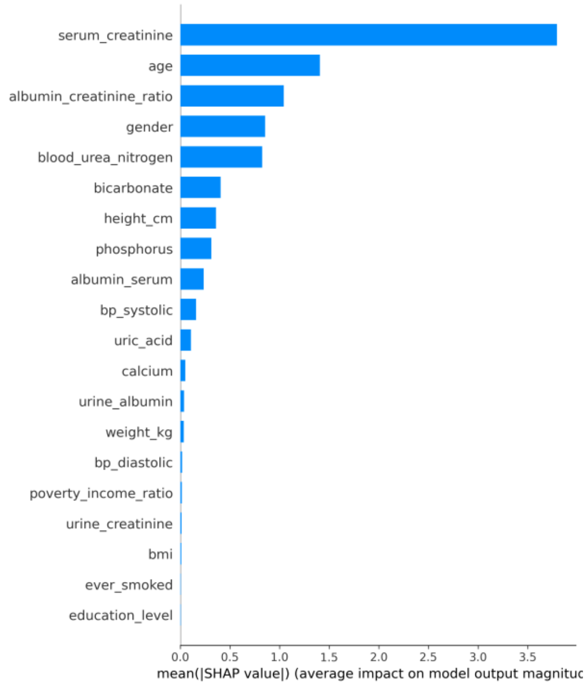
SHAP analysis of the NHANES-based XGBoost model revealed that Serum Creatinine was the most influential feature, followed by Age, Gender, Blood Urea Nitrogen, and Albumin-Creatinine Ratio. Higher values of Serum Creatinine, Blood Urea Nitrogen, and Age increased the likelihood of CKD prediction. These findings are clinically consistent with established biomarkers used in renal disease diagnosis, demonstrating that the model captures medically meaningful relationships rather than relying on spurious correlations.

SHAP Analysis on LightGBM — CKD Prediction

SHAP (SHapley Additive exPlanations) was applied to the trained LightGBM model to interpret feature contributions toward CKD prediction.

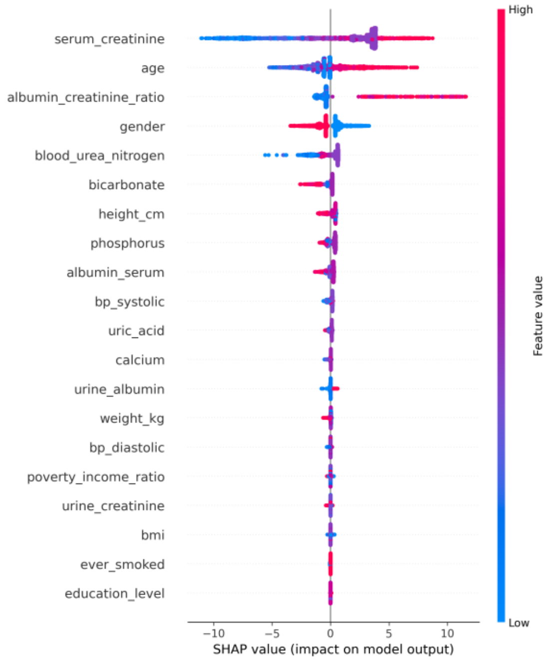
Global Feature Importance (Bar Plot)

serum_creatinine emerged as the single most dominant predictor with a mean SHAP value of ~3.8, far exceeding all other features. age and albumin_creatinine_ratio ranked second and third, followed by gender and blood_urea_nitrogen. Features like bmi, ever_smoked, and education_level contributed negligibly to the model's decisions.

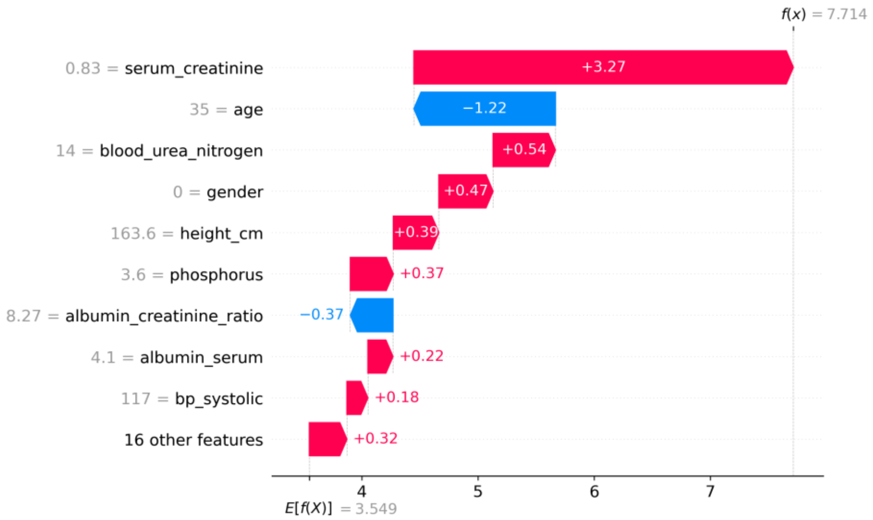


Feature Impact Direction (Beeswarm Plot)

- **Serum creatinine** — high values (red dots) pushed predictions strongly toward CKD, while low values (blue) pushed away, confirming its clinical role as the primary kidney function marker.
- **Age** — higher age consistently increased CKD risk, aligning with known epidemiology.
- **Albumin-creatinine ratio** — high values strongly increased CKD probability, which is expected as elevated urinary albumin is a hallmark of kidney damage.
- **Gender** — showed a directional split, suggesting one gender group carried higher predicted risk.
- **Blood urea nitrogen** — low values (blue) were associated with negative SHAP contributions, meaning normal BUN levels reduced CKD likelihood.



Single Prediction Explanation (Waterfall Plot)
For one sample patient (age 35, serum creatinine 0.83), the model predicted a score of **f(x) = 7.714** against a baseline of **E[f(X)] = 3.549**. The largest positive push came from serum_creatinine (+3.27), while young age acted as the main protective factor (−1.22). blood_urea_nitrogen, gender, and height_cm also contributed positively to the CKD prediction for this individual.



Overall Conclusion
The SHAP analysis confirms that the LightGBM model relies on clinically meaningful features — particularly renal biomarkers like serum creatinine, albumin-creatinine ratio, and BUN — rather than spurious correlations, which strengthens the model's trustworthiness for medical decision support.

Work on new UCI dataset(2023):

A newer Chronic Kidney Disease (CKD) dataset from the UCI Machine Learning Repository (2023 version) containing 202 records was explored and preprocessed. Multiple machine learning models, including Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree, Random Forest, AdaBoost, Gradient Boosting, and XGBoost, were trained and evaluated on this dataset.

During experimentation, all models achieved 100% classification accuracy and other performance metrics, indicating that the dataset was either highly separable or insufficiently challenging for meaningful model comparison. Since perfect performance across all models does not provide a realistic basis for evaluating model effectiveness and may indicate overfitting or dataset limitations, this dataset was not selected for the final study.

Consequently, the focus was shifted to more suitable datasets that provide a better assessment of model performance and allow meaningful comparison and interpretation of machine learning algorithms for CKD prediction.